

Jeonghoon Lim

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Education

- 2020-present 📖 **Iowa State University, Ames, IA**
Ph.D. in astrophysics: *Simulating the Streaming Instability and Planetesimal Formation*; Advisor: Prof. Jacob B. Simon
- 2018-2020 📖 **Chungnam National University, Daejeon, South Korea**
M.S. in astronomy: *Statistics of Turbulence Driven by Solenoidal and Compressive Drivings*; Advisor: Prof. Jungyeon Cho
- 2012-2018 📖 **Chungnam National University, Daejeon, South Korea**
B.S. in astronomy

Research Interests

- 📖 Planet formation theory, physics of protoplanetary disks, (magneto)hydrodynamic turbulence, computational astrophysics

Awards and Achievements

- 2022-2025 📖 Future Investigators in NASA Earth and Space Science and Technology (FINESST)
- 2022-2023 📖 Preparing Future Faculty Associate: Completed a two-semester professional development program offering advanced training in teaching, mentoring, and leadership to prepare graduate students for faculty careers
- 2020-2022 📖 The Korean Government Scholarship Program for Study Overseas
- 2020 📖 Graduate Research Award by Journal of the Korean Astronomical Society

Computational Skills/Experience

- 📖 Godunov Schemes; Particle-mesh; Grid codes
- 📖 High-performance computing on supercomputers in ACCESS and NASA-HEC
- 📖 Co-PI on 2022 – 2023 ACCESS Allocation (AST120062, 29 Million CPU Hours, PI: J. Simon, “Numerical Simulations of Planetesimal Formation”)
- 📖 Codes/Programming Languages: Athena, Julia, C, Python

Professional and Committee Services

- 2025 📖 External Reviewer for NASA’s Solar System Workings Program
- July 2018 📖 LOC Member, 8th East-Asia School and Workshop on Laboratory Space and Astrophysical Plasmas

Student Mentoring

Fall 2022 - Spring 2024  Olivia Brouillette, former undergrad at ISU; now in graduate program at New Mexico State University

Teaching Experience

Iowa State University, IA

Spring 2024  PHYS 132: General Physics II; Astro 150: Star, Galaxies, and Cosmology
Fall 2023  PHYS 115: Physics for the Life Sciences
Fall 2020  PHYS 111 Lab: General Physics

Chungnam National University, Daejeon, South Korea

Spring 2019  300-level Galactic Astronomy

References

Prof. Jacob B. Simon
Associate Professor
Iowa State University
jbsimon@iastate.edu

Prof. Wladimir Lyra
Associate Professor
New Mexico State University
wlyra@nmsu.edu

Prof. Steven Kawaler
Distinguished Professor
Iowa State University
sdk@iastate.edu

List of Publications and Products

As of Oct. 2025, **6** first-author articles • **9** total publications • **69** total citations on ADS (**64** from first-author papers) • **h-index: 4**

First-author articles

- 1 J. **Lim**, S. A. Baronett, J. B. Simon, C.-C. Yang, D. Sengupta, O. M. Umurhan, and W. Lyra, “Bridging Unstratified and Stratified Simulations of the Streaming Instability for $\tau_s = 0.1$ Grains,” *ApJ*, vol. 993, no. 1, p. 12, Nov. 2025, Art. no. 12.  DOI: 10.3847/1538-4357/ae01a6.
- 2 J. **Lim**, J. B. Simon, R. Li, O. Brouillette, D. G. Rea, and W. Lyra, “The Streaming Instability in 3D: Conditions for Strong Clumping,” *arXiv e-prints*, Sep. 2025.  DOI: 10.48550/arXiv.2509.18270. *arXiv*: 2509.18270.
- 3 J. **Lim**, J. B. Simon, R. Li, D. Carrera, S. A. Baronett, A. N. Youdin, W. Lyra, and C.-C. Yang, “Probing Conditions for Strong Clumping by the Streaming Instability: Small Dust Grains and Low Dust-to-gas Density Ratio,” *ApJ*, vol. 981, no. 2, p. 160, Mar. 2025, Art. no. 160.  DOI: 10.3847/1538-4357/adb311.
- 4 J. **Lim**, J. B. Simon, R. Li, P. J. Armitage, D. Carrera, W. Lyra, D. G. Rea, C.-C. Yang, and A. N. Youdin, “Streaming Instability and Turbulence: Conditions for Planetary Formation,” *ApJ*, vol. 969, no. 2, p. 130, Jul. 2024, Art. no. 130.  DOI: 10.3847/1538-4357/ad47a2.
- 5 J. **Lim** and J. Cho, “Decay of Turbulence in Fluids with Polytropic Equations of State,” *Journal of Korean Astronomical Society*, vol. 53, pp. 49–57, Apr. 2020.  DOI: 10.5303/JKAS.2020.53.2.49.
- 6 J. **Lim**, J. Cho, and H. Yoon, “Generation of Solenoidal Modes and Magnetic Fields in Turbulence Driven by Compressive Driving,” *ApJ*, vol. 893, no. 1, p. 75, Apr. 2020, Art. no. 75.  DOI: 10.3847/1538-4357/ab8066.

Second- or third-author articles

- 1 D. Carrera, L. E. J. Eriksson, J. **Lim**, W. Lyra, and J. B. Simon, “Positive feedback: II. How dust coagulation inside vortices can form planetesimals at low metallicity,” *A&A*, vol. 701, p. L1, Sep. 2025, Art. no. L1, **Contribution**: Participated in discussions of results and provided feedback on the manuscript.  DOI: 10.1051/0004-6361/202555045.
- 2 D. Carrera, J. **Lim**, L. E. J. Eriksson, W. Lyra, and J. B. Simon, “Positive feedback: How a synergy between the streaming instability and dust coagulation forms planetesimals,” *A&A*, vol. 696, p. L23, Apr. 2025, Art. no. L23, **Contribution**: Analyzed data from my streaming-instability simulations, participated in discussions of results, and provided feedback on the manuscript.  DOI: 10.1051/0004-6361/202554100.

Other co-authored articles

- 1 D. Godines, W. Lyra, L. Ricci, C.-C. Yang, J. B. Simon, J. **Lim**, and D. Carrera, “On the Mass Budget Problem of Protoplanetary Disks: Streaming Instability and Optically Thick Emission,” *arXiv e-prints*, Jun. 2025, **Contribution**: Participated in discussions of results and provided feedback on the manuscript.  DOI: 10.48550/arXiv.2506.10435.

Seminar & Conference Presentations

- 1 **Contributed talk**, Star and Planet Formation Group, University of Nevada, Las Vegas, NV, Sep. 2025.

- 2 **Contributed talk**, TCAN Disk Hydro & Planet Formation, University of Arizona, Tucson, AZ, Nov. 2023.
- 3 **Contributed talk**, Magnetic Fields in the Universe 7, Quy Nhon, Vietnam, Feb. 2020.
- 4 **Contributed talk**, Korean Astronomical Society Meeting, Seoul, South Korea, Oct. 2019.